The evolution of environmental impacts of platform chemicals linked to climate policies

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Introduction

Traditional production pathways of platform chemicals like ammonia and methanol are catalytic processes heavily dependent on fossil fuels, resulting in significant CO_2 emissions (560 Mt y^{-1}).[1] In recent years, emerging catalytic technologies have played a crucial role in advancing low-carbon production routes for these chemicals, with their environmental performance evaluated through life cycle assessments (LCAs).[2,3] However, the targets set by climate policies will necessitate substantial changes in the economy, including the decarbonization of energy production and transportation. Neglecting these future technological changes when assessing the environmental impacts of chemicals can result in misleading conclusions and guidance. Hence, in this work, we conduct a regionalized prospective life cycle assessment (pLCA) until 2050 to investigate how future economic trends linked to climate policies could impact the environmental footprint of catalytic chemical production routes.

Methods

We utilize the IMAGE[4] Integrated Assessment Model (IAM) to quantify the environmental impacts of fossil and green routes for ammonia and methanol production. This allows to project future economic scenarios and offers insights into the environmental impacts based on expected socio-economic and technological developments. Specifically, we adopt the intermediate shared socio-economic pathway (SSP2) and consider various representative concentration pathways (RCPs), including RCP6, RCP2.6, and RCP1.9, to generate a range of scenarios reflecting economic changes. To construct inventories for pLCAs spanning from 2020 to 2050, we employ the *premise*[4] (v1.5.9) framework. Furthermore, we expand the temporal resolution for the production of these chemicals to 26 regions worldwide, considering location-specific temporal profiles.

Results and Discussion

In contrast to fossil-based routes, the estimated climate change impacts of solar and wind-based ammonia production routes (taken herein to illustrate this study, **Figure 1a**) show a promising trend, with significant reductions anticipated by 2050. Under more ambitious climate policies, such as the 1.5 °C scenario, these impacts are expected to decrease substantially. Specifically, by 2050, both the solar and wind-based routes are projected to achieve almost a 70% reduction in impacts. In fact, low-carbon-based ammonia production routes exhibit the largest reductions in environmental impacts over time compared to hydrogen and methanol production. Furthermore, a regional analysis reveals that for solar (**Figure 1b**) and wind-based (**Figure 1c**) ammonia production, impact reductions vary significantly by region, emphasizing the importance of location-based dynamics for strategic planning.

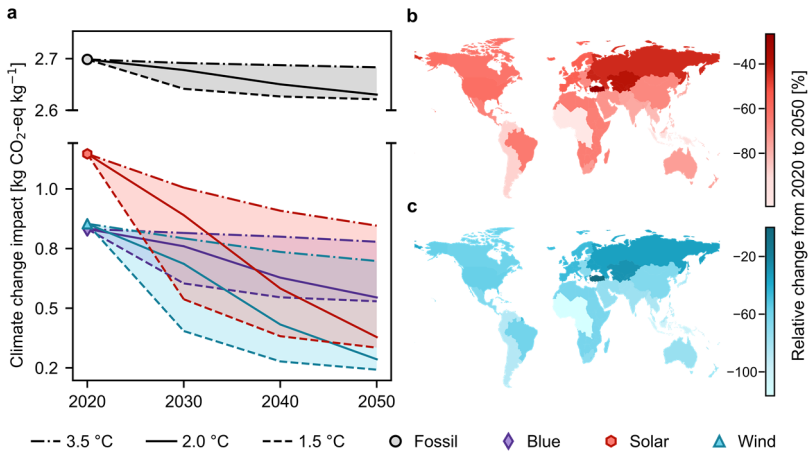

Figure 1. Climate change impacts of ammonia production pathways. (a) Global average impacts of fossil, blue, and green (both solar and wind-based) production from 2020 to 2050 across three climate policy scenarios (3.5 °C, 2 °C, and 1.5 °C). Regional analysis under the 2 °C scenario for (b) solar and (c) wind-based production.

Significance

Our results provide a more comprehensive outlook on the environmental performance of current and foreseeable catalytic routes for crucial platform chemicals. This assessment overcomes the traditional static analysis to provide valuable insights into the future, both on a temporal and geographical scale. These results equip policy making and technology investment with tools to navigate the evolving landscape of the chemical industry, ensuring informed and forward-thinking decisions.

References

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